

CSE 202 Midterm 3 Part (a)

Olcay Taner YILDIZ

I. QUESTION (HEAP) (15 POINTS)

Given the index of a heap node in a max-heap, write a recursive function that returns the sum of all descendants of this node.

```
double sumOfDescendants(int index)
```

II. QUESTION (DISJOINT SET) (15 POINTS)

Write a function that calculates the number of pair disjoint sets in a disjoint set structure. A disjoint set is pair, if the number of sets in that disjoint set is 2.

```
int numberOfPairs()
```

III. QUESTION (GRAPH) (15 POINTS)

Write a function that finds the average weight in a weighted graph. Write the function for both adjacency matrix and adjacency list representation.

```
int averageWeight()
```

IV. QUESTION (GRAPH) (15 POINTS)

Implement the graph data structure as an array of hash tables. Node i is connected to node j if the i 'th hash table contains a weighted edge j and its weight (for unweighted graphs edge weight is 1). Your implementation must contain the data structure, constructor function, two addEdge functions for unweighted and weighted graphs respectively. You must also implement hash function for the Edge class.